



Improving [REDACTED] Perfect Order Rate

Natalie Huynh
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Agenda



1

About Me



2

Project Scope



3

Next Steps



4

Internship Reflection



About Me





About Me



College: UC San Diego

Major: Math-Computer Science

Minors: Data Science, Cognitive Science

Hobbies:

- Kayaking
- Traveling
- Cooking + Eating



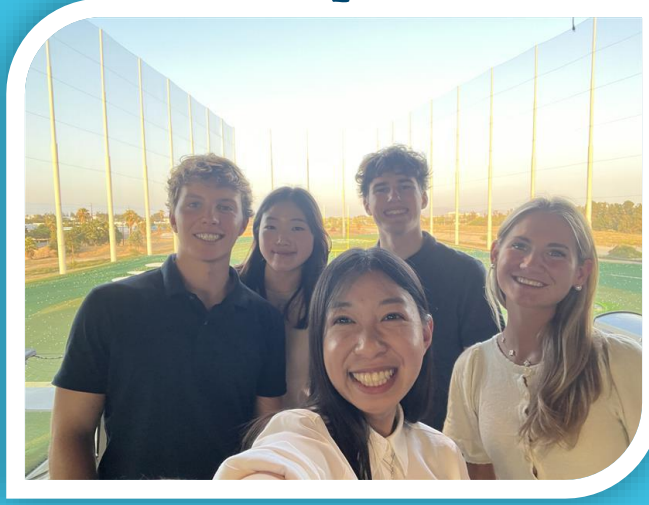


Onboarding + Plant Tour

Rialto Plant Tour



SIP



Welcome Week





Project Overview: Sam's Club DCs Perfect Orders

1

Introduction to
[redacted] DCs
and Perfect
Order process

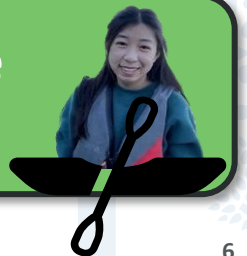
2

Meeting with
stakeholders
to understand
order process
for [redacted]

3

Analyzing data
to derive
perfect order
insights

Objective: Root cause highest imperfect activity and identify opportunities to improve touchless order processing for [redacted]





Project: Orders

DC's: 8% of Sam's Total Order Volume

Context

Service Level: 89.6%

2024 vs 2025 Orders

2024

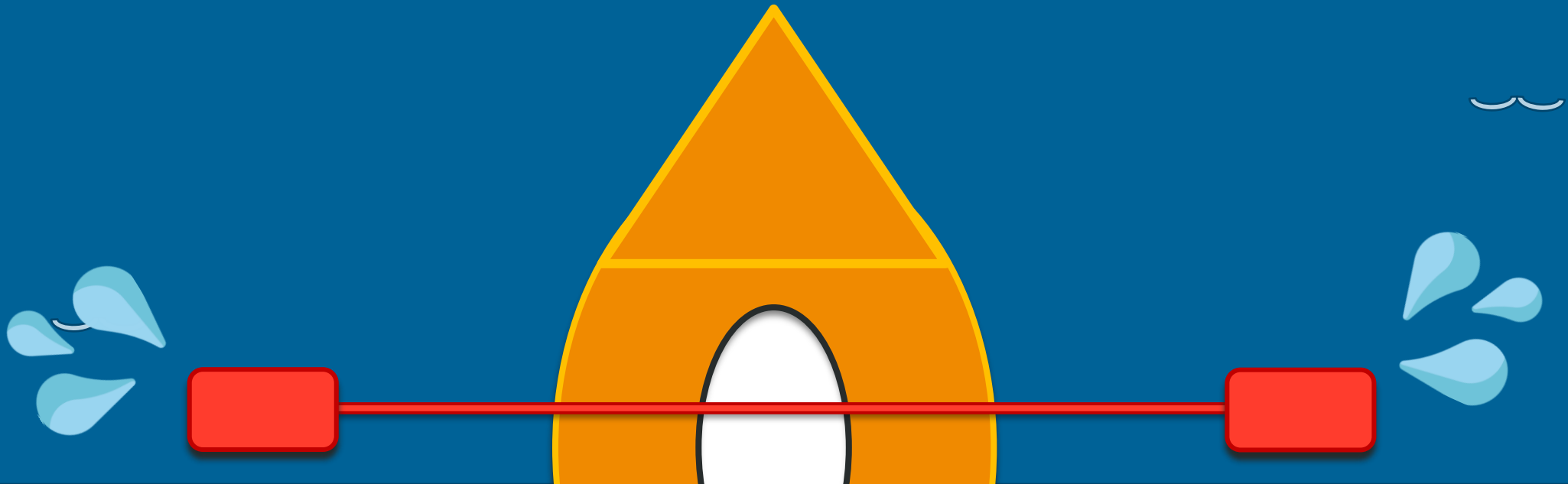
Sales Orders:

2025 YTD

Sales Orders:



Project Scope





Defining Perfect Orders

Perfect order was created as a consolidated metric for the Order to Cash process.



Quantity delivered is same as
initial sales order

Default ORG on order entry is
the same as ship-from

Order delivery
is not delayed



Date
Change

Quantity
Change

SKU
Change

ORG
Change

Canceled
Orders



Delivery appointment only
scheduled once

SKU delivered is same as one
on the initial order

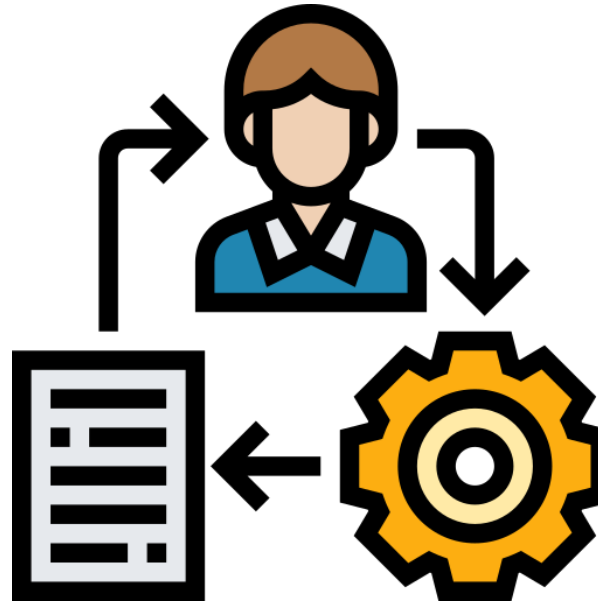
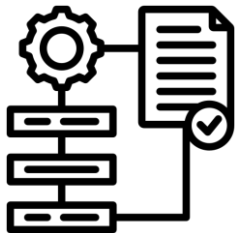
Order is not
canceled



Methodology and Approach

Collaborate with Stakeholders

Outline Solutions



Clean, Filter, Process Data

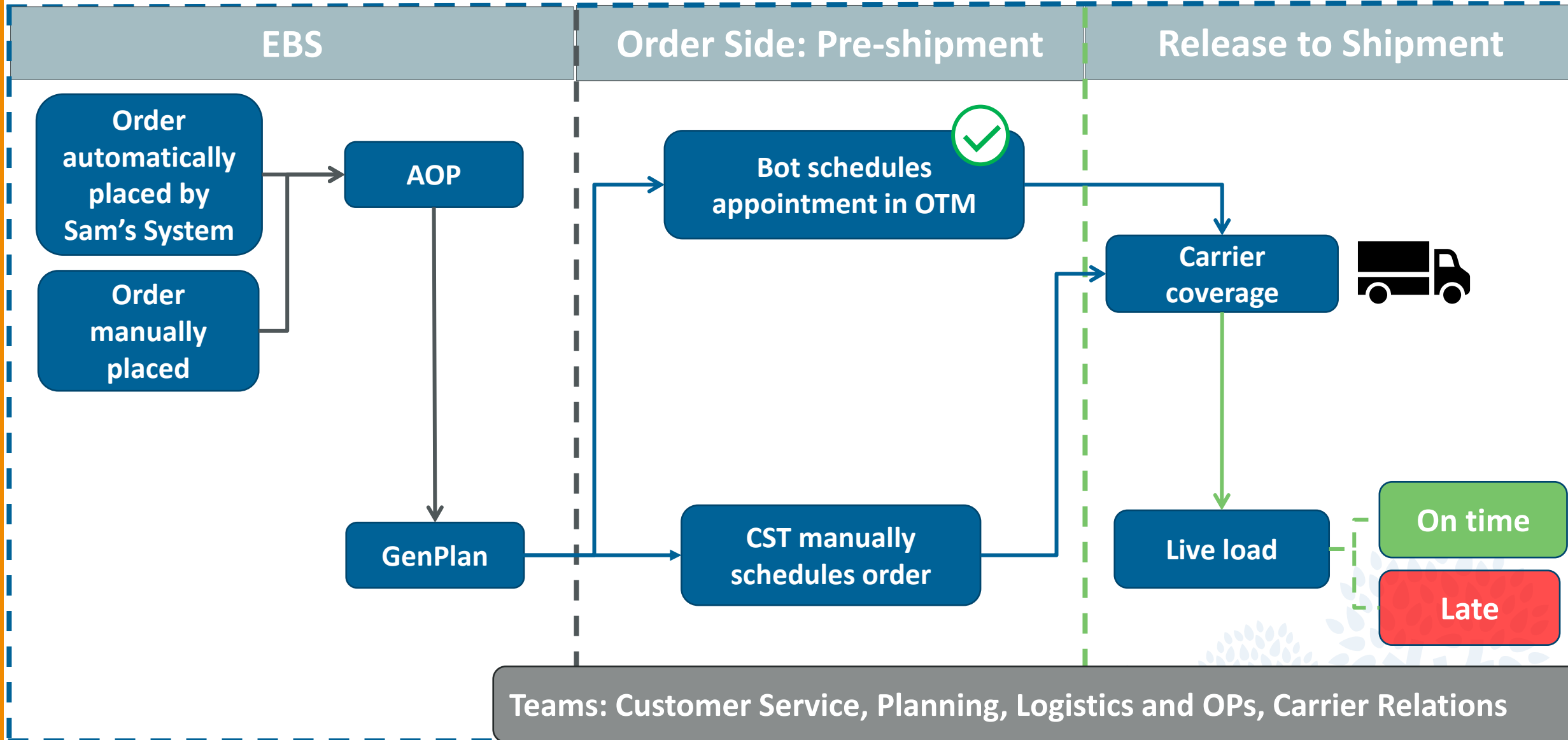


pandas

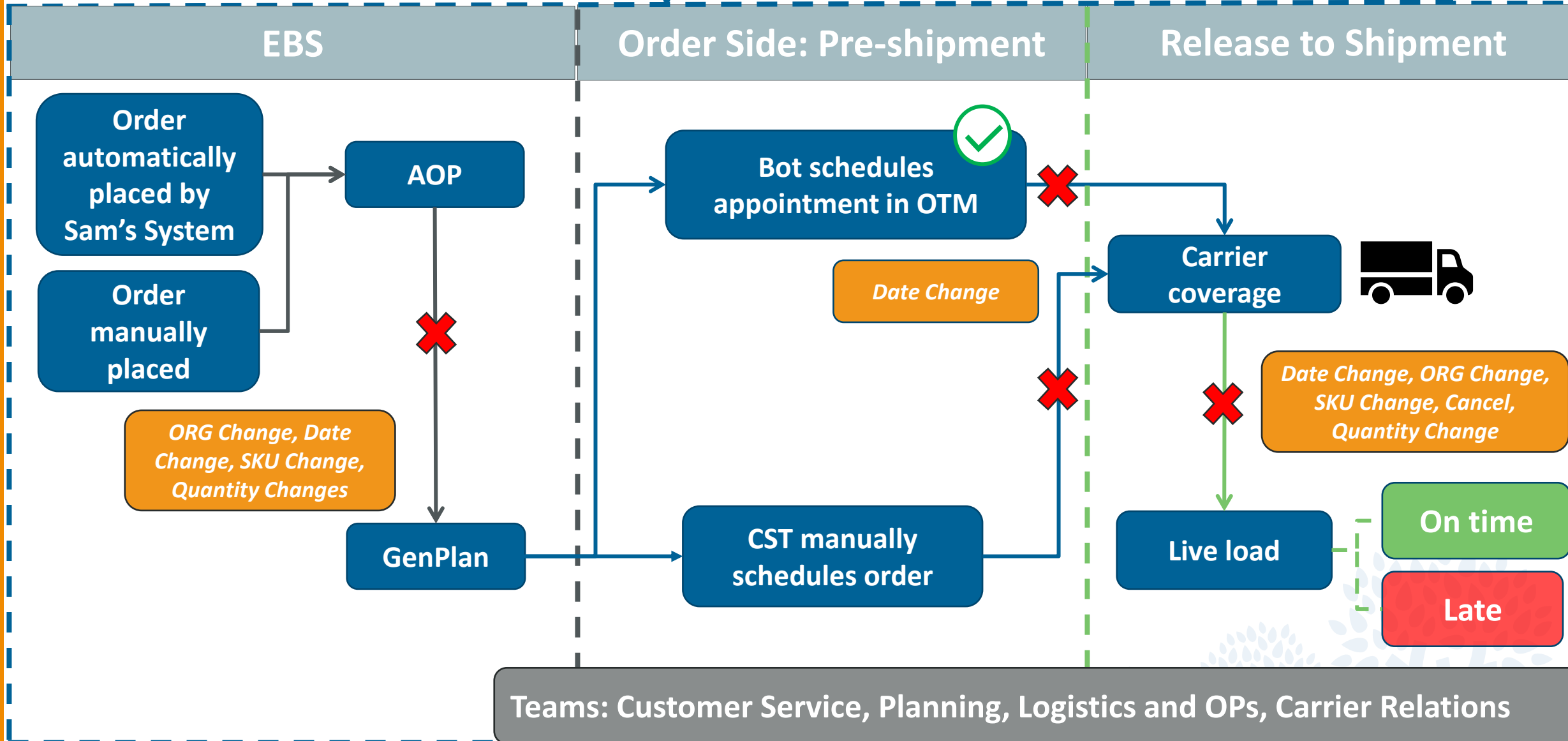




Order Process Flow:

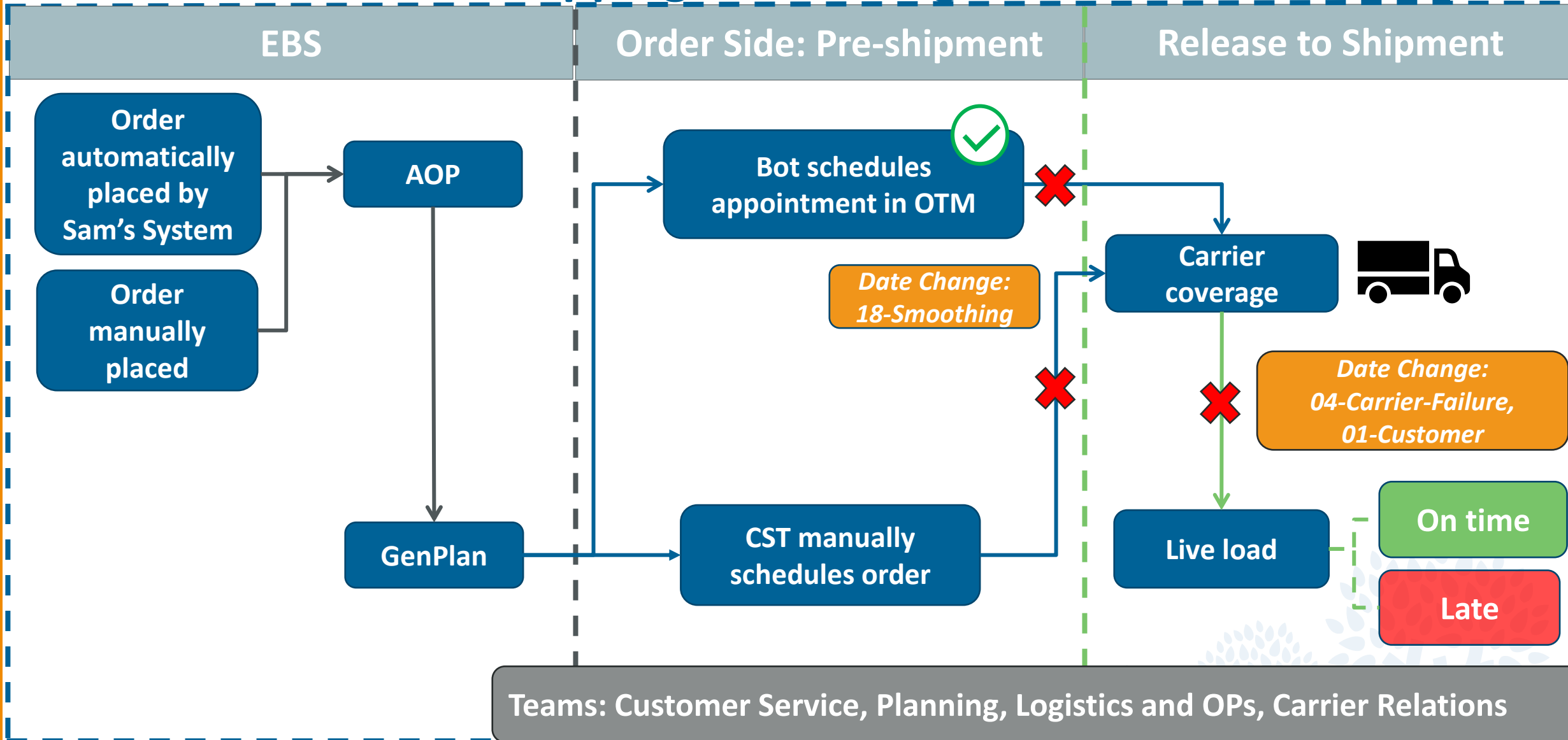


Order Process Flow + All Imperfect Activities:





Order Process Flow Mapping Date Change:





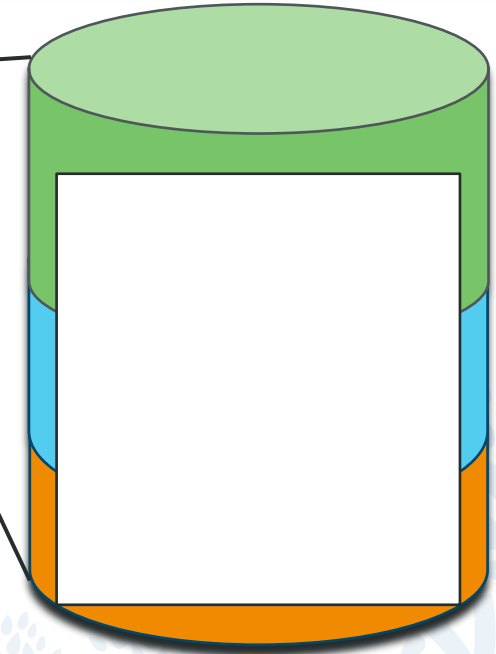
Imperfect Order Overview

Which bucket has the highest impact on Perfect Order rate?

Date Changes

Categories	# of Orders Impacted	% of Orders Impacted	Cost
Date Change			
ORG Change			
Cancelled Orders			
SKU Change			
Quantity Change			

Reason Codes





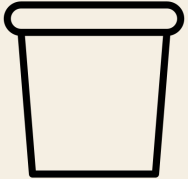
Imperfect Activities and Service Level

Do our imperfect activities allow us to deliver on time?

Date
Change



Imperfect
Activities



Imperfect Bucket	% Met Original Request Date	Total Orders	Total Orders Date Changed
ORG Change			
SKU Change			
Quantity Change			

Date changes are frequently associated with other imperfect activities.



Date Changes: 18-Customer-Volume Smoothing



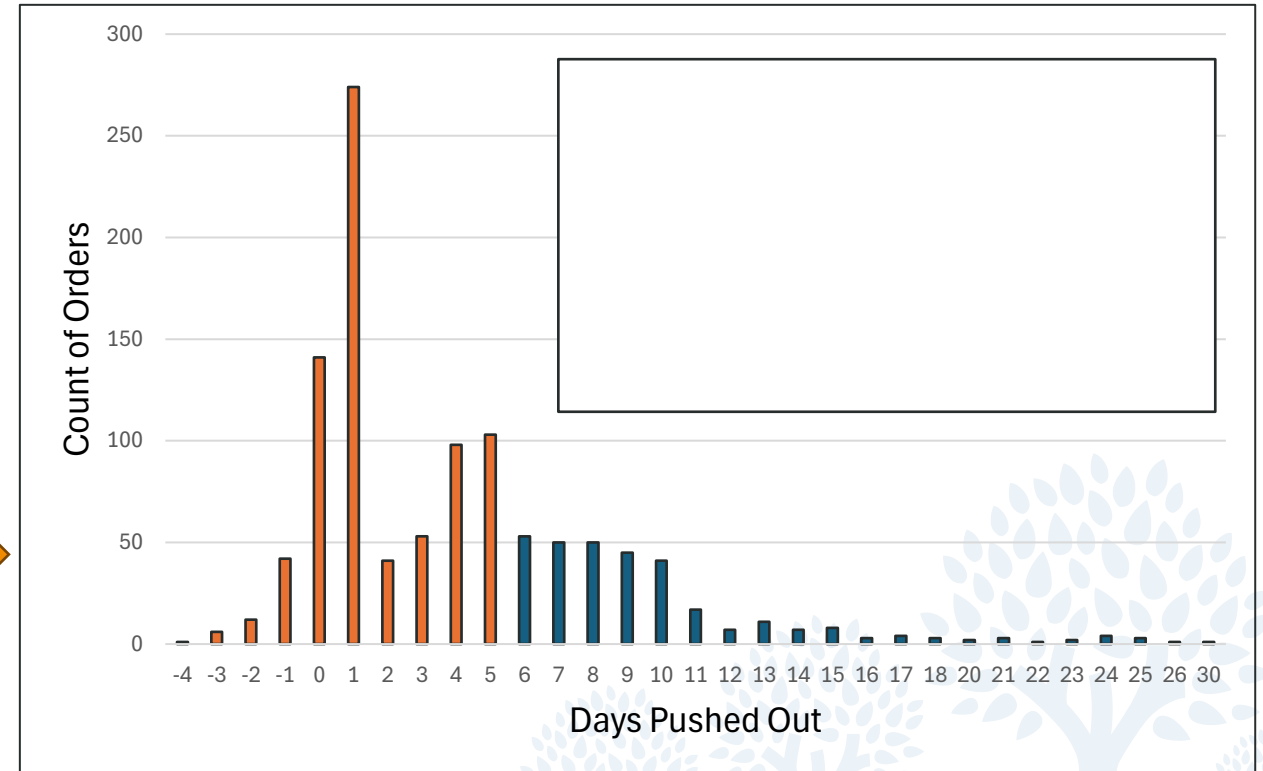
Smoothing Definition: Customer orders over forecasted amount. Delivery dates are adjusted to production capacity.

Customer Service and Planning Feedback

- No restriction on order volume
- Orders are manually created to help with increase in demand
- **1.5%** of orders are manually created
- Orders are smoothed based on receiving appointment availability



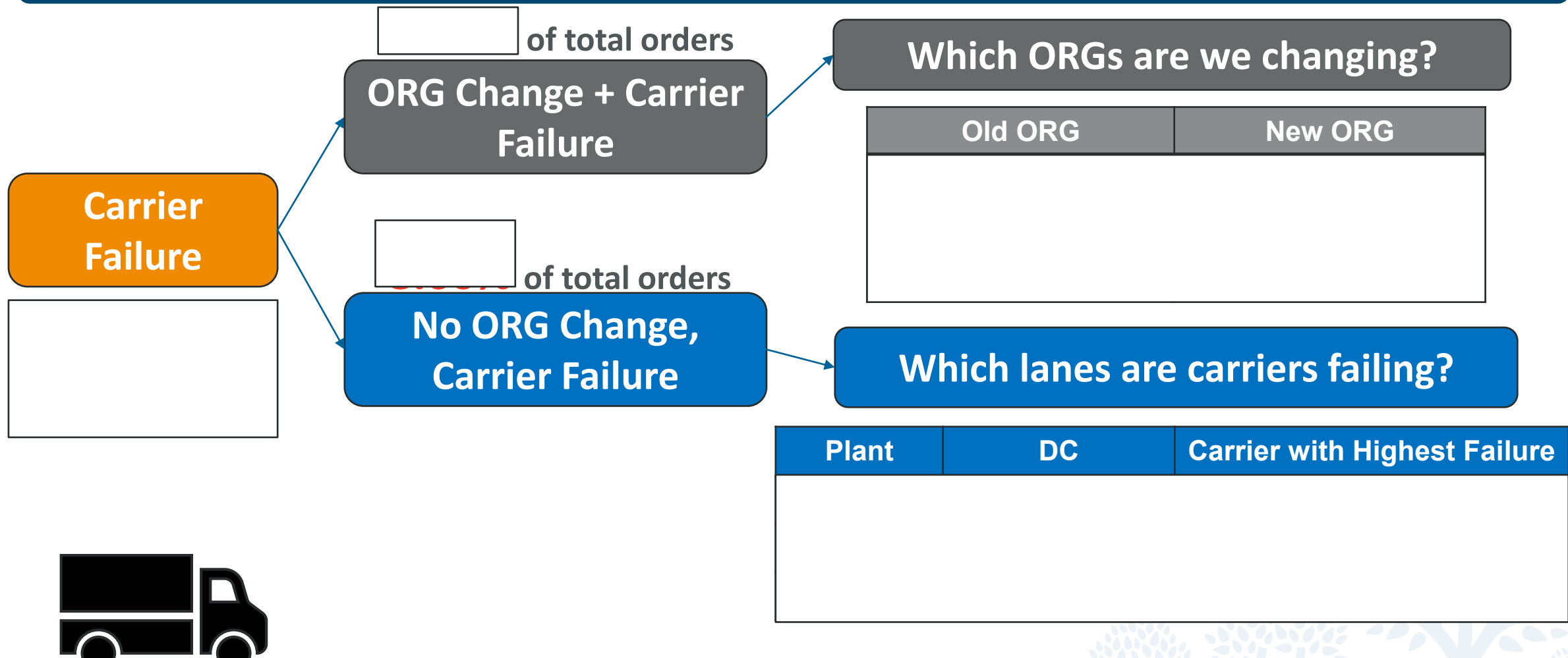
How far out are orders smoothed?



Date Change : 04-Carrier Failure



 **Carrier Failure Definition:** Carrier fails to pick up an order as scheduled which can result in order delivery delays.

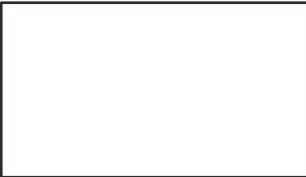




Date Changes: 01-Customer-Customer



01-Customer-Customer Definition: Customer reaches out if they want to change delivery date of an order.



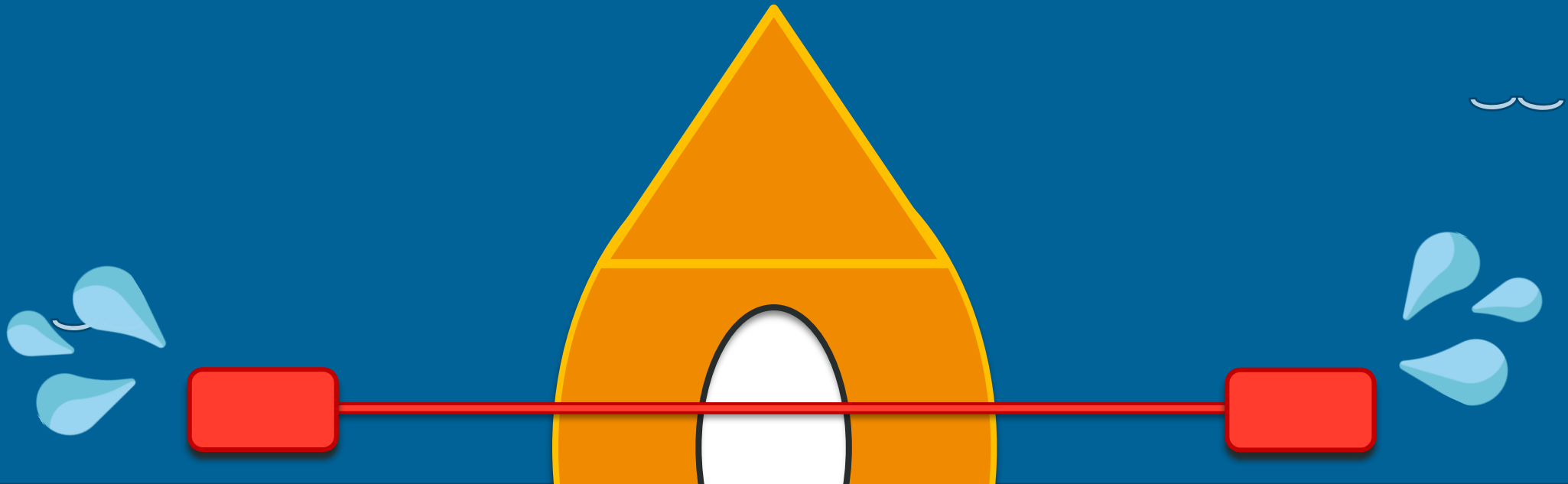
Avg. Days Pushed Out:

Avg. Days Pulled Up:

Top Requesting Order Changes

Club Location	Count of Orders	Push Out	Pull Up

Next Steps





Next Steps: Driving Solutions

Date Change

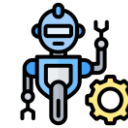
Solution

Order Smoothing:

- Leverage automated Ordering System in place of manual appointments
- Request appointment slot for Niagara deliveries

Impact to Imperfect Order Rate

- ↓ 2% for a cost savings of **\$15,400**



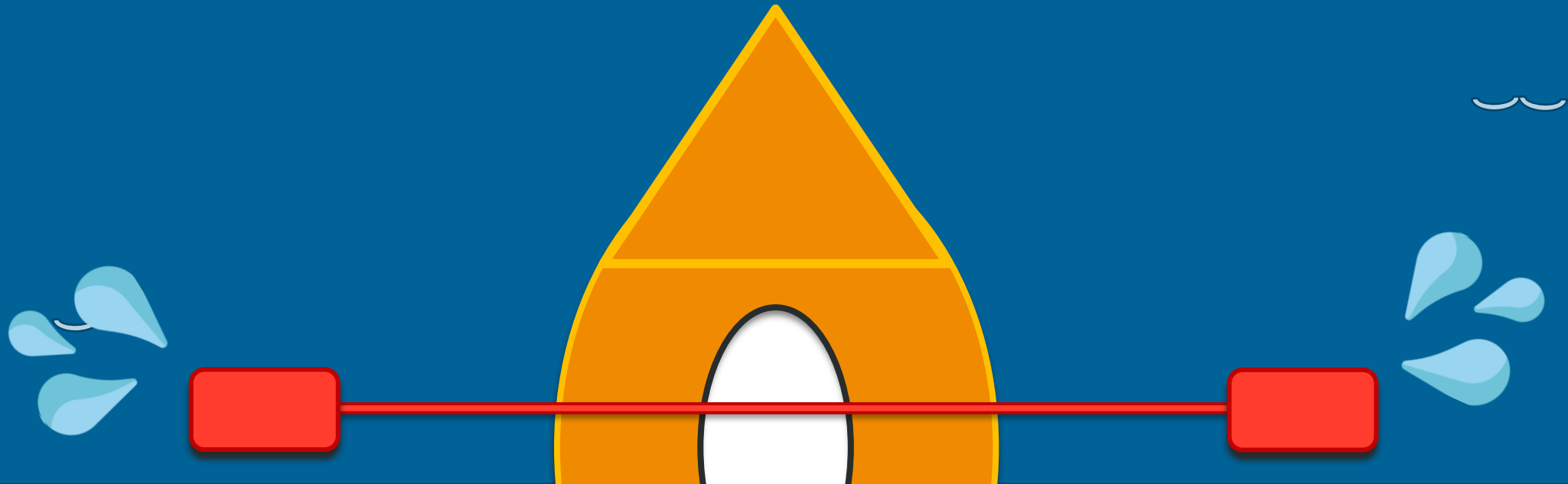
- ↓ 11% for a cost savings of **\$89,952**



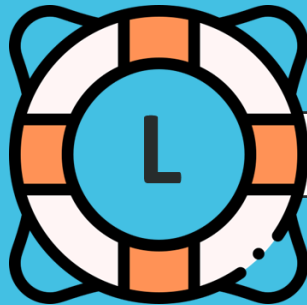
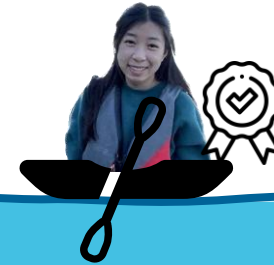
Total Projected Annual Cost Savings: \$104,152



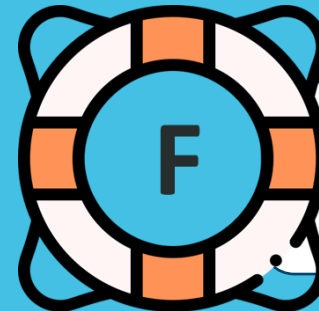
Internship Reflection



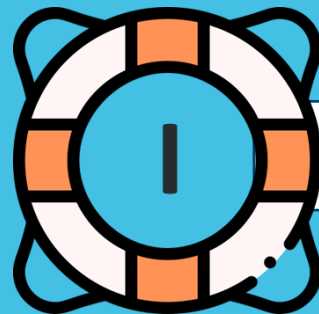
Takeaways: Niagara LIFEsavers



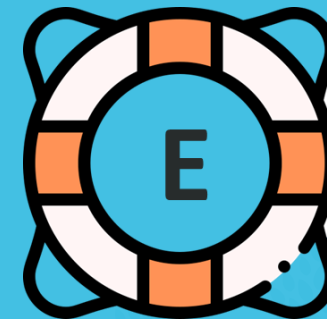
Stakeholder Alignment



Collaborate and ask for Feedback



Experiment + Pivot



Communicate Ideas and Insights





Internship Reflection

Career Development:

- SIP Conference
- BT & DDI Coffee Chats
- Rotational Info Sessions (ITRP, SCALE, PDP)



Skills Learned:

- Navigating meetings with cross-functional teams
- Process Mapping
- Data Cleaning and Analysis (ETL)
- Presenting Insights





Thank you!

Annie Chang
Shounak Deo
Michelle Ortiz
Karla Gonzalez

BT Team
Stella Mooraj
Talent Acquisition
Talent Development

Richard Chavez
Katie Lento
Gina Alison
Mariah Minera
Darlene Coronado
Julian Dean
Lisa Drewitz
Rebecca Owen
Julie Ceja
Tequila Baxter
Layne Sutherland



Questions?

Data Appendix

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The diagram consists of two rectangular boxes. The top box is white with a black border and contains the text "Carrier Failure with ORG Changes". A blue arrow points from the left side of this box towards the left edge of the slide. The bottom box is also white with a black border and contains the text "Carrier Failure with No ORG Changes". An orange arrow points downwards from the bottom center of this box.

Carrier Failure with ORG Changes

Carrier Failure with No ORG Changes



27



Imperfect Activities and Service Level: RS Code

Imperfect Bucket	% Met Original Request Date	Total Orders	Total Orders Date Changed	Reason Code

